# TEST PLAN

## MediVerify – Medicine Authenticity Checker

Prepared by: Muhammad Ali Khan | Date: October 2026

## 1. Introduction

MediVerify is a web application that helps users check whether a medicine is genuine or fake. The user can either upload an image of the back of the medicine or type the medicine code. The app then shows the medicine name, expiry date and whether it is legit or fake. It also provides sign up, login, a history of previously verified medicines, and an About page.

Application URL: https://mediverify-check.vercel.app

## 2. Objective

To check that the main features of MediVerify (registration, login, medicine verification by code and by image, result display, history and navigation) work correctly, and to find and report defects before real users face them.

## 3. Scope

**In scope:**

- Login and Sign Up (valid, invalid and empty inputs)
- Medicine verification by code and by image upload
- Result display: medicine name, expiry date, legit/fake status
- Verification history, user sessions and logout
- About page, navigation links, basic UI and browser compatibility

**Out of scope:**

- Performance and load testing
- Full security penetration testing (only basic input checks are covered)
- Accuracy and completeness of the medicine database

## 4. Types of Testing

- **Functional testing:** checking that each feature works as expected.
- **Negative testing:** wrong, empty or unexpected inputs (wrong password, fake code, wrong file type).
- **Boundary testing:** very long inputs and large files.
- **UI and compatibility testing:** mobile screen, different browsers, page loading.
- **Basic security testing:** basic checks such as SQL injection text and access to history after logout.

## 5. Test Environment

| Item | Details |
|---|---|
| Application | https://mediverify-check.vercel.app (deployed web app) |
| Browsers | Google Chrome, Microsoft Edge, Mozilla Firefox |
| Devices | Desktop/laptop and mobile screen view |
| Test case tool | Microsoft Excel / Google Sheets |
| Defect tracking | Bug report files stored on GitHub |
| Version control | GitHub repository: mediverify-qa-testing |

## 6. Test Data

- A code of a genuine medicine and a random fake code
- Clear, blurry, cropped and non-medicine images (JPG and PNG)
- An unsupported file (PDF/TXT) and a very large image
- An expired medicine, valid and invalid email addresses, strong and weak passwords

## 7. Test Approach

Test cases were written module by module with an ID, steps, test data and expected result. Each test case was then executed manually on the live application, the actual result was recorded, and the status was marked as Success or Failed. Every failed test case is reported as a separate bug with steps to reproduce, expected and actual result, severity and a screenshot.

## 8. Entry and Exit Criteria

**Entry criteria:**

- The application is deployed and reachable
- Test cases and test data are ready

**Exit criteria:**

- All 50 test cases are executed
- Every failed test case is logged as a bug report

## 9. Test Deliverables

- Test Plan (this document)
- Test Cases (50 cases in Excel/CSV)
- Bug Reports with screenshots
- README with project summary

## 10. Risks and Assumptions

- The application may be slow or unavailable because it is hosted online
- Test data is limited, so not every real medicine could be tried
- Some expected results are based on the intended behavior of the app

## 11. Test Execution Summary

All 50 planned test cases were executed. 37 passed and 13 failed. Most failures are in Code Verification, Result Display and About / Navigation.

| Module | Total | Success | Failed |
|---|---|---|---|
| Login | 7 | 7 | 0 |
| Sign Up | 7 | 7 | 0 |
| Code Verification | 8 | 3 | 5 |
| Image Upload | 8 | 7 | 1 |
| Result Display | 4 | 2 | 2 |
| History | 6 | 5 | 1 |
| About / Navigation | 4 | 1 | 3 |
| Session | 3 | 2 | 1 |
| UI / Compatibility | 3 | 3 | 0 |
| **Total** | **50** | **37** | **13** |